

La dimensión del espacio tangente coincide con la de la variedad

Corolario 1. Sea M una variedad diferenciable de dimensión n

$$\implies \forall p \in M : \dim T_p M = n \text{ (como espacio vectorial).}$$

Demostración: Sea (U, ψ) una carta de M que contiene a p , tenemos

$$T_p M \xleftarrow{D\iota|_p} T_p U \xrightarrow{D\psi|_p} T_{\hat{p}} \hat{U} \xrightarrow{D\iota'|_{\hat{p}}} T_{\hat{p}} \mathbb{R}^n$$

donde $D\psi|_p$ es un isomorfismo (propiedad prop-direfencial-apl-diferenciable/?? de prop-direfencial-apl-diferenciable/??) y $D\iota|_p$ y $D\iota'|_{\hat{p}}$ son isomorfismos (prop-esp-tangente-abie Por tanto, $\dim T_p M = \dim T_{\hat{p}} \mathbb{R}^n = n$ ya que $T_{\hat{p}} \mathbb{R}^n \cong \mathbb{R}^n$ (por el teo-esp-tangente-rn-isomorfo-rn/??)

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Referenciado en

- cor-variedades-difeomorfismos-misma-dimension

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